QPS TRIAGE · ABACUS · CODEX · FEDERATED DMAIC

NOT LIFTED · GATE NOT RESTORED

Global Code Purple

A recursively self-testing execution protocol across repositories, agents, CI, PR admission and evidence. It builds on QPS TRIAGE, ABACUS, CODEX, KEB and DOW and adds no second authority.

STATUS AT	GitHub readback 2026-10-18 13:27 UTC · not lifted: 2 of 5 lift criteria met
SEED	CODEX main · federation/mesh/CODE_PURPLE_EXECUTION_SEED_20261012.yaml
OWNER	GBO · Process Engineer, QPLANT System Owner, SCK CEN

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EXECUTIVE SUMMARY

ABACUS has merged nothing since 6 October; one PR unblocks it

WHAT IS BLOCKED

All 20 open ABACUS PRs. The **Main Hard Gate** ruleset requires one check, `PR Feedback Gate`, which waits for an exact-head Codex code and security review. The Codex allowance is exhausted, so the gate never turns green.

THE FIX IN FLIGHT

ABACUS #1858 lets the owner attest the exact head when Codex cannot run. It cannot approve itself: the gate runs main's code. It needs a one-time owner **break-glass** on the ruleset to land.

ALREADY MERGED

CODEX #984 (this protocol's seed) merged at 12:16 with 4 red checks. CODEX main has no required-check ruleset, so nothing stopped it. CODEX main is green again on its current head.

Decisions needed from GBO: (1) accept the 4 inherited reds under break-glass (they fail nightly on main) · (2) add the one-time bypass on ruleset 24494381. Done: threads resolved, #1878/#1879 opened, policy-and-inventory fixed (e7616e62).

READBACK

Tracked heads at 17:55

ITEM	SUBJECT	HEAD	STATE	PASS / FAIL / NEUTRAL	RED CHECKS
ABACUS #1858	Owner-attested review fallback	e7616e62	open	87 / 2 / 1	Code Quality, Security Scanning (Coverage, Full Red Proof still running) · policy-and-inventory now green
ABACUS #1859	PR-level new-debt ratchet	6fac58cc	open	61 / 4 / 1	Code Quality, Full Red Proof, Security Scanning, validate-scripts
ABACUS #1857	Fix syntax error in diagnostic_kpi_probe.py	e7b45912	open	60 / 4 / 1	Code Coverage, Code Quality, Full Red Proof, Security Scanning
ABACUS #1856	Lossless branch hygiene burn-down	f11d8044	open	91 / 3 / 1	Code Coverage, Code Quality, Full Red Proof
CODEX #984	Code Purple execution seed	90e3396a	merged 12:16	55 / 4 / 0	ruff, governance-gate ×2, validate (at merge)
CODEX main	after #984 merge	768c9e35	main	38 / 2 / 0	validate (fixed on main at f9ae4676), osv-scan (fails on every main push since before #984)
CODEX main	current head	cce9ce20	main	—	MissionControl CI green; OSV-Scanner red (pre-existing)

Counts exclude skipped and cancelled runs. ABACUS main's last merge is #1838 (2026-10-06).

TAXONOMY

Four kinds of reference, one model

STRUCTURE · WHAT THE PROTOCOL DOES

1–8 Stages
The pulse written as a procedure. Each stage groups nodes and lists entry, actions and exit.

contain ↓

N01–N15 Nodes
One step of a pulse: who owns it, which artefact, what evidence it writes. Drawn on the pulse graph.

are joined by ↓

E01–E19 Edges
Typed connections: sequence, pass, red, hold, evidence, recursion.

LIFECYCLE · WHAT MOVES A WORK ITEM

12 States
Where a PR head or red finding is: test pending, set-up, run, debug, fix, integrate, admission, follow-up, hold, merged, federate.

are changed by ↓

T01–T16 Triggers
One event, one guard, one action. Each trigger names the nodes it acts at and whether it is wired today.

act at → Nodes

EVIDENCE · WHAT PROVES IT

per node Telemetry
Preconfig (stage, test, debug), run attempts with pass / fail / neutral / pending, errors, logs.

is kept as ↓

sha Receipts & census
Exact-commit receipts per run; one census row per iteration N.

is learned as ↓

REX Lessons
Recurring causes and reusable fixes, read back at the start of the next pulse.

Worked example. Trigger **T11** moved #1858 from *Admission* to *HOLD*. It acts at nodes **N10** (gate) and **N11** (hold), which belong to stages **6** and **7**. On the pulse graph that move is edge **E12**, and the hold record is evidence for REX.

Phase letters **D M A I C** (Define, Measure, Analyse, Improve, Control) tag every node and stage.

LEGEND

Colour, line and status vocabulary

NODE ROLES (FILL)

Authority
Resolves or owns state: controller, allocation, output

Process
Checks and verification

Gate
A binary decision

Repair
Isolated change and retest

Admitted path
Merge and post-merge proof

Hold
Blocked, logged, retried

Evidence store
REX and KEB

EDGE TYPES (LINE)

Sequence
Normal next step

Pass
Admissible successor

Material red
Goes to repair

Hold
Deferred or blocked

Evidence
Receipts and lessons

.self recursion
Output N becomes input N+1

MARKERS

Receipt
Step writes an exact-commit receipt

Run bar
pass · fail · pending share of attempts

Tracked PR
Where a live PR is now

IMPLEMENTATION STATUS

Seeded / committed

In progress

Partial

On hold

Not implemented

Not reached

TRIGGER WIRING

Wired — Fires automatically today

Partly wired — Detection automatic, response manual

Manual — A person does it

Proposed — Not built yet

GLOSSARY

Terms used in this deck

DMAIC	Define, Measure, Analyse, Improve, Control: the phases each node belongs to.
.self	The recursive interface: validated output N becomes input N+1 through governed reconciliation (⊕).
Material red	A completed failing check that blocks admission, as opposed to neutral or pending.
Break-glass	A one-time, owner-only bypass of the ruleset, restored immediately and recorded in REX.
Pre-match	Reusing a result already proven for the identical commit and inputs instead of re-running it.
REX · KEB	CODEX evidence stores: reusable lessons and recurrence (REX), evidence identity (KEB).
GLOB · MESH_DOCK	Global capability census and the producer/consumer contract between repos.
SIPOC	Supplier, Input, Process, Output, Customer. With .self, the customer of N supplies N+1.
Exact head	The one commit SHA under test. Evidence for any other SHA does not count.
Non-waivable gate	A required check or review that must be green on the same SHA before merge.
Single-flight	One admitted PR merges at a time so no head goes stale behind another merge.
Receipt	A small record binding a result to an exact commit and input hash.
DOW	ABACUS derived analysis: ranks findings and returns scoped dispositions.
Inherited debt	A red already present on main, as opposed to a defect the PR introduced.

THE CENTRAL RULE

Seven steps, in order, every pulse

01

Reuse proven evidence

D→M

02

Allocate, preconfigure, pretest

M

03

Find the first material red

A

04

Repair the smallest common cause

I

05

Verify independently

I

06

Publish exact-head evidence

C

07

Advance only admissible successors

C

A passing local preflight raises confidence for the next iteration. Only exact-head GitHub checks and reviews move a PR through admission.

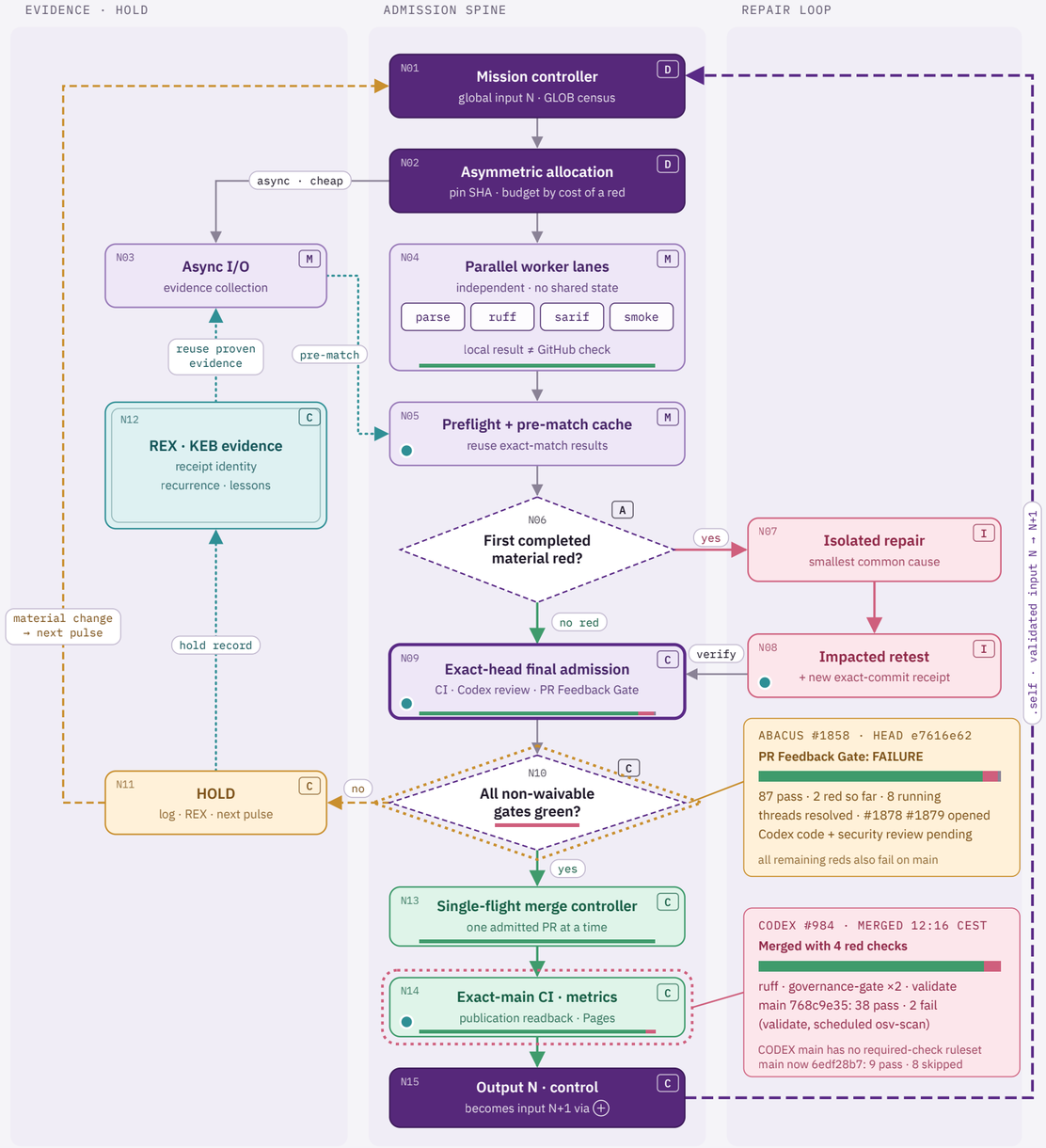
EXECUTION PULSE

One pulse, three lanes

The admission spine runs down the centre. Reds go right into the repair loop; holds and evidence sit on the left. The dotted rings mark where the two tracked PRs are now.

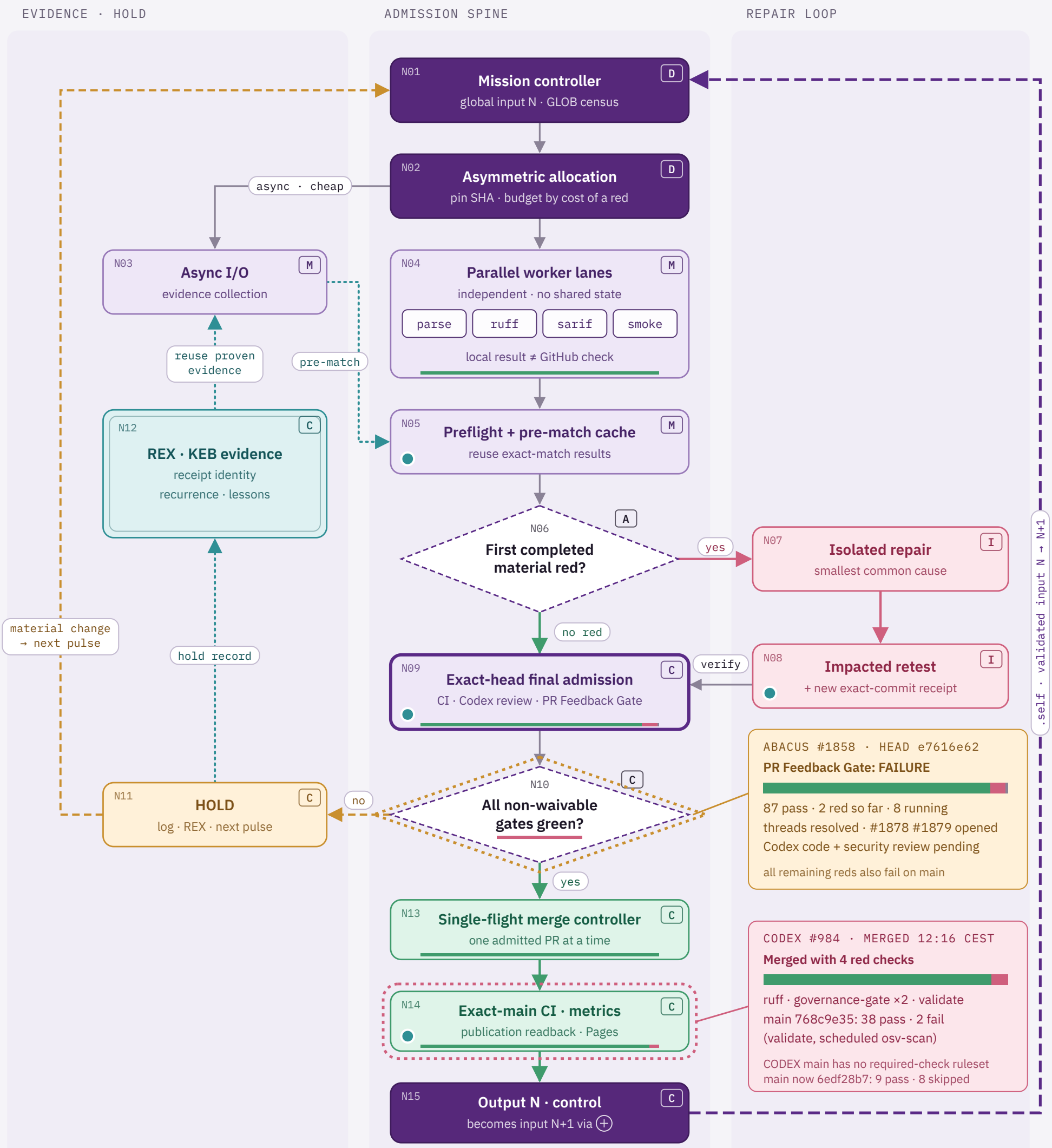
- Authority
- Gate
- Admitted path
- Evidence store
- Process
- Repair
- Hold

#1858 waits at N10 (gate). **#984** reached N14 without passing N10.



EXECUTION PULSE · PORTRAIT DETAIL OF SLIDE 09

One pulse, three lanes



Authority · Process · Gate · Repair
Admitted path · Hold · Evidence store · Sequence
Pass · Material red · Hold · Evidence
.self recursion

N01	Mission controller	Define	N02	Asymmetric allocation	Define
N03	Async I/O	Measure	N04	Parallel worker lanes	Measure
N05	Preflight + pre-match cache	Measure	N06	First completed material red?	Analyse
N07	Isolated repair	Improve	N08	Impacted retest	Improve
N09	Exact-head final admission	Control	N10	All non-waivable gates green?	Control
N11	HOLD	Control	N12	REX · KEB evidence	Control
N13	Single-flight merge controller	Control	N14	Exact-main CI · metrics	Control
N15	Output N · control	Control			

PROCESS STAGES 1–4 · CLICK A NODE ID FOR ITS REFERENCE ROW

From trigger to first red

1 DEFINE — Mobilise

N01 Mission controller N02 Asymmetric allocation

ENTRY
A trigger: new head, schedule or material change

ACTIONS
Census authorities and capabilities
Pin the exact SHA
Budget lanes by the cost of a red

EXIT
Pinned SHA pair
Lane budget

2 MEASURE — Collect and run

N03 Async I/O N04 Parallel worker lanes

ENTRY
SHA pinned, lanes budgeted

ACTIONS
Collect prior evidence asynchronously from REX, KEB and GitHub
Run parse, Ruff, SARIF and smoke lanes in parallel

EXIT
Evidence bundle
Per-lane results

3 MEASURE — Pre-match

N05 Preflight + pre-match cache

ENTRY
Lane results and evidence bundle

ACTIONS
Reuse results proven for the identical commit and inputs
Run only the remainder

EXIT
Preflight receipt (local, not CI proof)

4 ANALYSE — Triage

N06 First completed material red?

ENTRY
Preflight receipt

ACTIONS
Rank findings with DOW
Assign one of four failure classes
Select the first completed material red

EXIT
One red to repair, or none

PROCESS STAGES 5–8 · CLICK A NODE ID FOR ITS REFERENCE ROW

From repair to the next pulse

5 IMPROVE — Repair loop

N07 Isolated repair · N08 Impacted retest

ENTRY
One classified red

ACTIONS
Fix the smallest common cause on an isolated branch
Retest the impacted set
Write a new exact-commit receipt

EXIT
Repaired head submitted for verification

6 CONTROL — Admission

N09 Exact-head final admission · N10 All non-waivable gates green?

ENTRY
Exact head with no known material red

ACTIONS
Run CI, Codex review and PR Feedback Gate on the exact SHA
Evaluate non-waivable gates only

EXIT
Admit, or hold

7 CONTROL — Hold and learn

N11 HOLD · N12 REX · KEB evidence

ENTRY
A non-waivable gate red or pending

ACTIONS
Record the reason in REX
Update recurrence and lessons
Plan the next pulse

EXIT
Hold record
Next pulse on material change

8 CONTROL — Merge and prove

N13 Single-flight merge controller · N14 Exact-main CI · metrics · N15 Output N · control

ENTRY
Admission green

ACTIONS
Merge one PR at a time
Run exact-main CI, record metrics, read back the publication
Write the census row

EXIT
Output N becomes input N+1

NODE REFERENCE 1 / 2 · STAGE AND TRIGGER IDS ARE LINKS

Mobilise, measure, triage, repair

ID	NODE · WHAT IT DOES	ROLE	STAGE	TRIGGERS	ARTEFACT	STATUS
N01	**Mission controller** Takes global input N, discovers authorities and capabilities, and resolves producer, consumer, receipts and scope for this pulse.	Authority	1	T01 T07 T12	CODEX GLOB.yaml, .mesh/MESH_DOCK.yaml	Seeded / committed
N02	**Asymmetric allocation** Pins the exact head under test and spends compute where a red is most expensive. Cheap I/O runs async; costly checks get dedicated lanes.	Authority	1	T01	Seed YAML execution contract	Seeded / committed
N03	**Async I/O** Collects existing evidence without blocking the workers, including prior receipts that may already prove a result.	Process	2	T02	REX and KEB lookups, GitHub check readback	Not implemented
N04	**Parallel worker lanes** Runs parse, Ruff, SARIF security scans and smoke in parallel with no shared state.	Process	2	T02	ABACUS scripts/testpilot_preflight.py, pre-commit, Ruff, Bandit, Semgrep	Partial
N05	**Preflight + pre-match cache** Merges worker output with collected evidence. A result already proven for the identical commit and inputs is reused instead of rerun.	Process	3	T03	Receipts keyed by exact commit and input hash	Not implemented
N06	**First completed material red?** Ranks findings and selects the first completed red that matters, classified as source defect, inherited debt, review capacity or runner failure.	Gate	4	T03 T03b T04	ABACUS DOW	Not implemented
N07	**Isolated repair** Fixes the smallest common cause only. Security-sensitive governance changes cannot approve themselves.	Repair	5	T05	Smallest-scope branch	Not implemented
N08	**Impacted retest** Reruns only what the repair touches and writes a new exact-commit receipt.	Repair	5	T06	Impacted test set	Not implemented

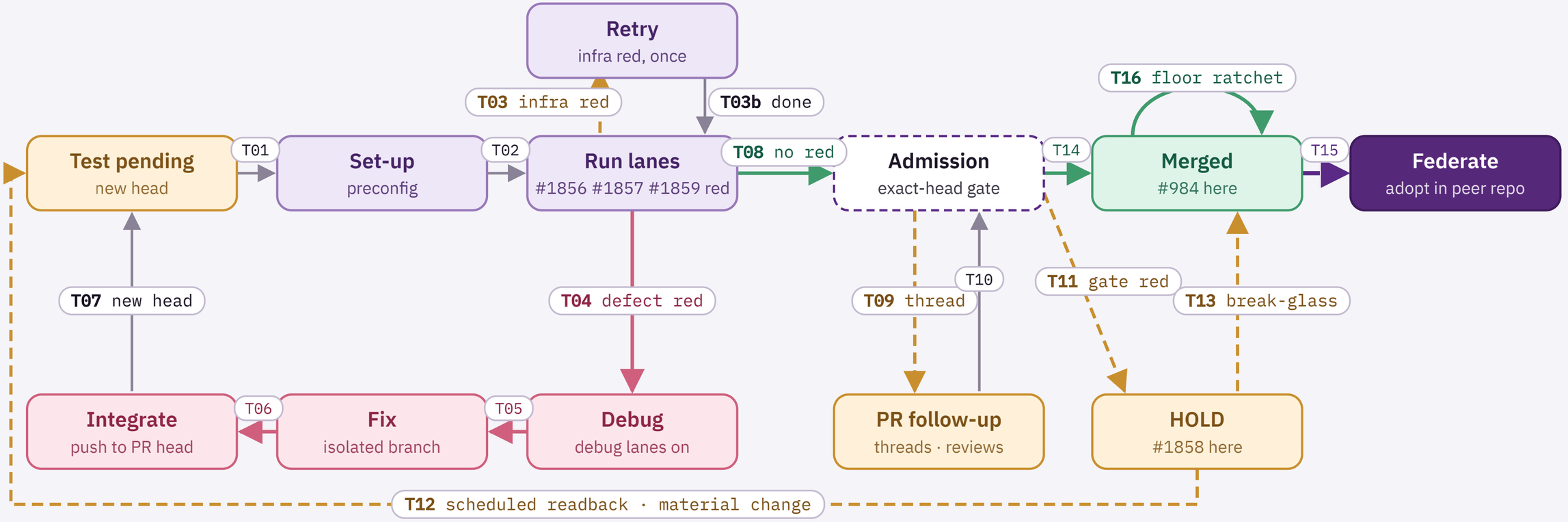
NODE REFERENCE 2 / 2 · STAGE AND TRIGGER IDS ARE LINKS

Admit, hold, learn, merge, prove

ID	NODE · WHAT IT DOES	ROLE	STAGE	TRIGGERS	ARTEFACT	STATUS
N09	**Exact-head final admission** Independent verification on the exact head SHA, including required reviews.	Process	6	T08 T09	GitHub Actions, PR Feedback Gate, Codex review and security review	In progress
N10	**All non-waivable gates green?** Admits only when every non-waivable check and review is green on the same SHA. Waivable checks are recorded, not counted.	Gate	6	T10 T11 T13	Required checks and reviews at exact head	On hold
N11	**HOLD** Logs the blocking reason to REX and plans the next pulse. A material change re-enters at the mission controller.	Hold	7	T09 T11 T12	REX entry	On hold
N12	**REX · KEB evidence** Holds evidence identity, recurrence counts and reusable lessons. Async I/O reads from it at the start of each pulse.	Evidence store	7	T11 T16	CODEX KEB and REX	Partial
N13	**Single-flight merge controller** Merges one admitted PR at a time so no head goes stale behind another merge.	Admitted path	8	T13 T14	Merge queue discipline	Not reached
N14	**Exact-main CI · metrics** Proves main is green after the merge, records metrics and reads the published result back.	Admitted path	8	T15 T16	GitHub Actions on main, Pages	Not reached
N15	**Output N · control** Becomes the input of iteration N+1 through governed reconciliation (⊕).	Authority	8	T15	Iteration census	Not reached

LIFECYCLE

What moves a work item from state to state



T01 T02 T03 T03b T04 T05 T06 T07 T08 T09 T10 T11 T12 T13 T14 T15 T16

Each arrow is a trigger T01–T16: one event, one guard, one action. Tracked PRs: #1858 sits in HOLD (via T11), #984 in Merged (via T14 without a gate).

TRIGGERS 1 / 2

Set-up, run, debug, fix, integrate

ID	FROM → TO	TRIGGER EVENT	ACTION	NODES	WIRED TODAY
T01	Test pending → Set-up	Pull request opened or synchronised, or a push	Start CI; mark the gate pending	N01 N02	Partly wired GitHub starts CI on these events. ABACUS's gate re-evaluates on workflow_run and review events, not on synchronize (#1858 thread, line 599).
T02	Set-up → Run lanes	Preconfig receipt written	Start the parallel lanes	N03 N04	Proposed Needs the .self harness.
T03	Run lanes → Retry	A lane goes red with class runner or infrastructure (5xx, runner lost, cancelled)	Re-run the failed jobs once; log a REX entry	N05 N06	Manual Re-run all jobs button on GitHub.
T03b	Retry → Run lanes	Retry finished	Results re-enter triage	N06	Manual Same re-run.
T04	Run lanes → Debug	Red classified as a source defect that doesn't reproduce locally	Re-run that lane with debug on (debugpy, LLDB) and keep the trace	N06	Partly wired CODEX main already runs LLDB, lldb-dap and debugpy lanes (green); nothing triggers them from a red yet.
T05	Debug → Fix	Root cause recorded	Branch from the exact head	N07	Manual —
T06	Fix → Integrate	Impacted retest green, with an exact-commit receipt	Push to the PR head	N08	Manual —
T07	Integrate → Test pending	New head on the PR	Old receipts retired; the cycle restarts at T01	N01	Wired Push event.
T08	Run lanes → Admission	All lanes terminal	Exact-head CI plus Codex code and security review	N09	Partly wired CI is automatic. The Codex review request is manual, and the allowance is exhausted.

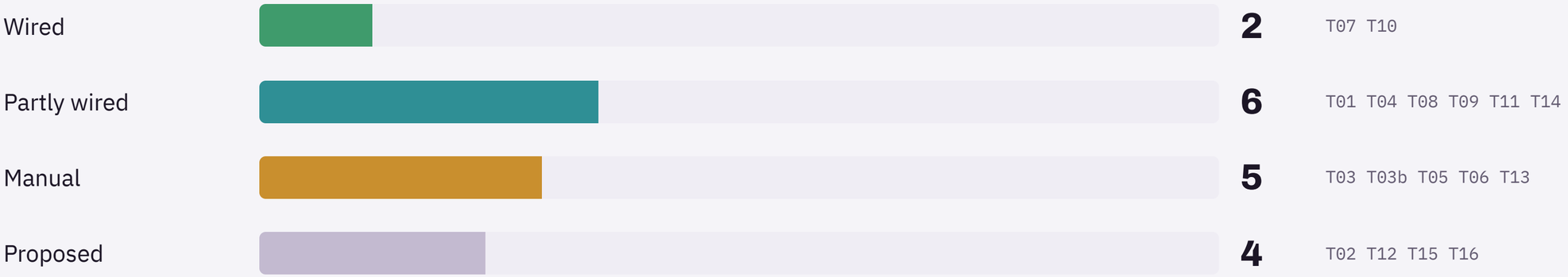
TRIGGERS 2 / 2

Admission, follow-up, hold, merge, federate

ID	FROM → TO	TRIGGER EVENT	ACTION	NODES	WIRED TODAY
T09	Admission → PR follow-up	Review thread, Codex comment or gate failure	Answer, fix, or record an owner residual-risk decision	N09 N11	Partly wired Detection is wired: pr-feedback-gate.yml runs on pull_request_review and review-comment events. The response is manual.
T10	PR follow-up → Admission	Thread resolved or review submitted	The gate re-evaluates	N10	Wired Review events trigger the gate.
T11	Admission → HOLD	A non-waivable gate is red or pending (e.g. Codex outage)	Write the HOLD record and a REX entry	N10 N11 N12	Partly wired The gate holds. The REX record is manual.
T12	HOLD → Test pending	Scheduled readback, or a material change	Start a new pulse	N11 N01	Proposed Needs a scheduled task (burndown item 7).
T13	HOLD → Merged	Owner decision with a written residual risk	Squash-merge, restore the ruleset, write REX	N10 N13	Manual Owner edits ruleset 24494381. This is the proposed path for #1858.
T14	Admission → Merged	All non-waivable gates green on the same SHA	Squash-merge	N13	Partly wired ABACUS: ruleset enforces it. CODEX: no ruleset; #984 merged with 4 reds.
T15	Merged → Federate	Exact-main CI green and read back	Open an adoption PR in the peer repo; write the census row	N14 N15	Proposed Not built. For #984, CODEX main has 2 post-merge reds to clear first.
T16	Merged → Merged	Exact-main CI measures coverage at or above floor + 1 point	Raise the coverage floor to the measured value; write the census row (total, breadth, depth, per area)	N14 N12	Proposed Not built. Today the gate is a fixed 75% and fails on every PR.

AUTOMATION COVERAGE

Only 2 of 17 transitions fire on their own



AUTOMATIC TODAY

Fully automatic: a new head restarts the cycle (T07) and review events re-run the gate (T10). Partly automatic: CI starts on push and PR events but the gate does not reset on synchronize (T01), and thread detection is wired while the response is manual (T09).

NEXT TO WIRE

The `.self` harness (T02 set-up receipt), a scheduled readback pulse (T12), and debug-on-red using the existing CODEX LLDB and debugpy lanes (T04).

ABACUS #1858 · HEAD E7616E62

Gate invariants: 2 met, 3 failing, 1 running

✗	**Exact-head Codex code review complete**	Pending; Codex allowance exhausted
✗	**Exact-head Codex security review complete**	Pending; Codex allowance exhausted
✓	**Zero unresolved review threads**	Both resolved 17:20 as owner residual risk → #1878, #1879
✗	**Current-head GitHub Actions terminal and green**	Code Quality and Security Scanning red; all inherited from main (see next slide)
…	**Current-head admission evidence terminal and green**	Gate re-running on e7616e62
✓	**Known deferred warnings linked to open issues**	#1736

The gate checks out the default branch, so #1858's own fallback cannot admit #1858. The code reviews stay unmet until Codex is back or the owner uses break-glass.

FAILURE CLASSIFICATION

Four reds fail on main every night: inherited, not caused by PRs

CHECK	MAIN NIGHTLY	#1858	#1859	#1857	#1856	CLASS
Code Coverage	red	running		red	red	Inherited debt · confirmed
Code Quality Checks	red	red	red	red	red	Inherited debt · confirmed
Full Red Proof Summary	red	running	red	red	red	Inherited debt · confirmed
Security Scanning	red	red	red	red		Inherited debt · confirmed
policy-and-inventory		green				Source defect · fixed in e7616e62
validate-scripts			red			Unclassified

POLICY-AND-INVENTORY ON #1858 · FIXED

`docs/ci/WORKFLOW_RATIONALIZATION.md` went stale after the one-line policy change. Regenerated and pushed as e7616e62; the check is green on the new head.

RECURRING REDS · CONFIRMED ON MAIN

ABACUS main (53688f26) fails Security Scanning, Code Quality, Code Coverage and Full Red Proof in every nightly CI/CD Test Suite run since at least 7 October. Each is an “Enforce … red roots / threshold” step: inherited debt. #1859 targets exactly this.

LANDING #1858

Break-glass sequence, owner and Claude steps

1	Claude	Done: threads resolved, #1878 #1879 opened and corrected	17:20 · 17:55
2	Claude	Done: regenerated rationalization doc pushed	e7616e62 · green
3	GBO	Accept the 4 inherited reds under break-glass	they fail nightly on main
4	GBO	Add yourself as bypass actor on ruleset 24494381	one-time
5	Claude	Squash-merge #1858, read back ABACUS main CI	merge receipt
6	GBO	Remove the bypass immediately	ruleset restored
7	Claude	Write the REX entry; re-evaluate the 20 open PRs	owner attestation now available

#1878 and #1879 turned out to be fixed in the PR already (a7f623b5, 14430046); they stay open as post-merge checks. Ruleset at 17:55: active, 0 bypass actors.

CODEX #984

Merged, but the admission step was skipped

WHAT HAPPENED

#984 merged at 12:16 with 4 red checks on head 90e3396a: ruff, governance-gate ×2, validate. CODEX main has no required-check ruleset, so the merge went through. After the merge, 768c9e35 had 38 pass and 2 fail. **validate** (deterministic materializations) was fixed on main at f9ae4676. **osv-scan** fails on every main push, including before #984: inherited.

WHY IT MATTERS FOR CODE PURPLE

The protocol's own rule (advance only admissible successors) was not enforced on the repository that holds the protocol. ABACUS enforces a gate that is too strict to pass; CODEX enforces none.

Decision for GBO by 14 October: give CODEX main a required-check ruleset like ABACUS, or record why not.

COVERAGE · MEASURED ON #1858 HEAD 203D63AF, 1,086 TESTS

Coverage is 37%: breadth × depth

OVERALL

36.9%

line + branch, as the gate measures it

LINES

38.2%

10,686 of 27,998 statements

BREADTH

64.5%

statements in files any test touches (155 of 269 files)

DEPTH

59.2%

lines covered inside those touched files

```
coverage = breadth × depth = 0.645 × 0.592 = 0.38 → target 0.90 × 0.84 ≈ 0.75
```

WHY IT READS ~30%

CI recorded 37% on 1 October over a wider scope (41,408 statements). 114 files have no test at all; they hold 9,952 statements, 57% of the gap.

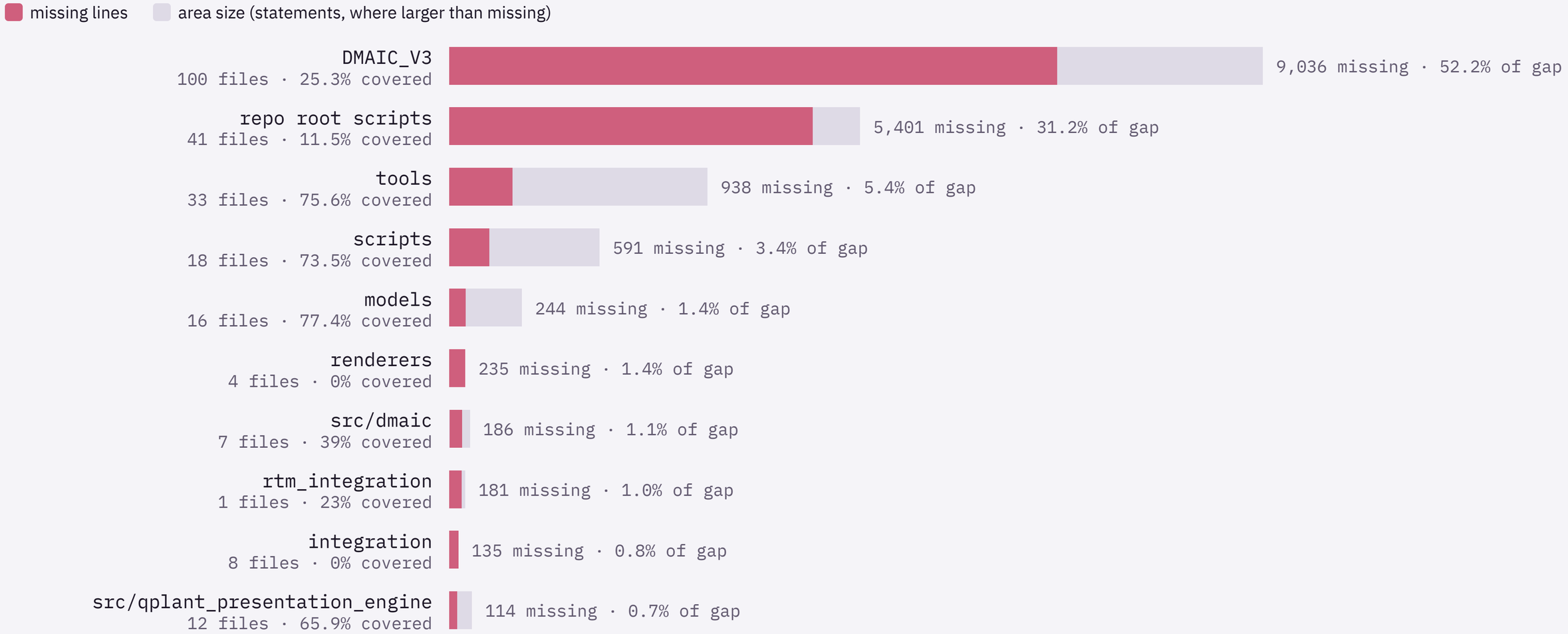
WHAT THE GATE DOES TODAY

`Enforce coverage threshold` fails at a fixed 75% on every PR, so the same red appears everywhere and steers nothing.

Local run with the CI command (`pytest --cov=.`, branch on). Docker integration tests deselected (no Docker here); CoolProp tests skip as in CI.

COVERAGE HIERARCHY · REPO → AREA → FILE

Two areas hold 83% of the 17,312 missing lines



COVERAGE · WHAT IS DONE TODAY

Measured on every PR, but nothing ratchets it upward

IN PLACE

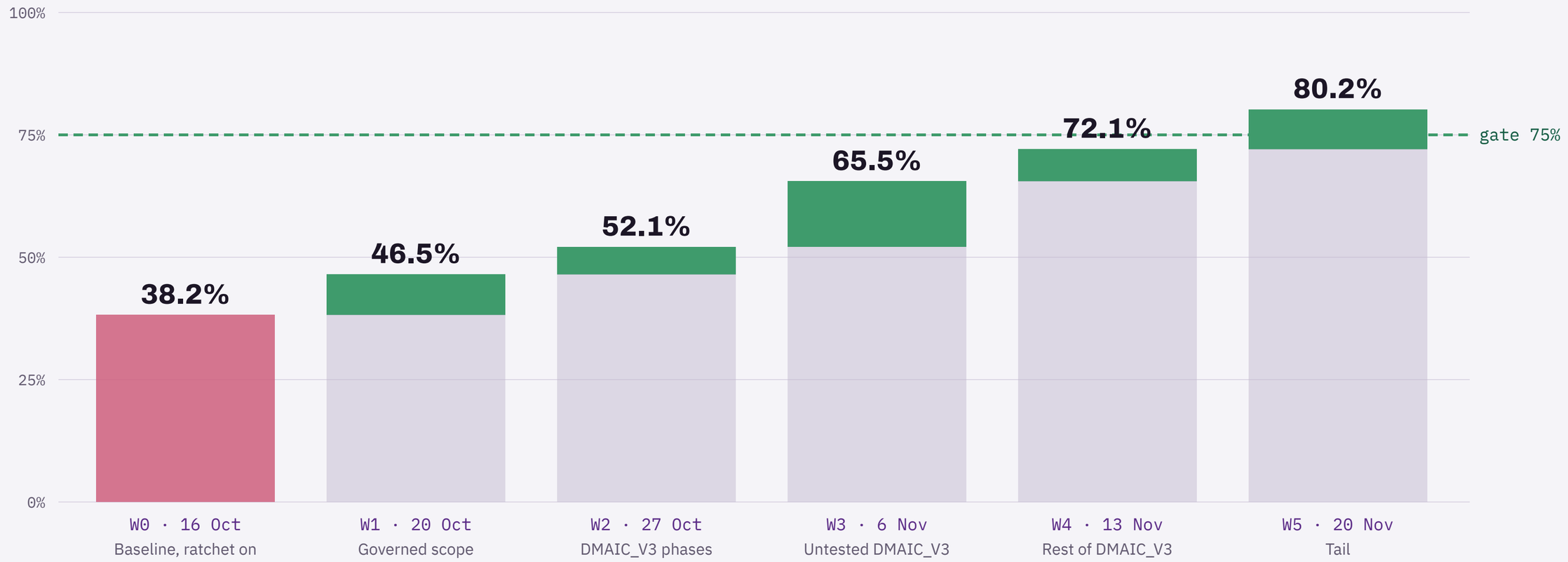
- `ci-cd-tests.yml` measures broad coverage on every PR and writes it to the job summary
- A ranked debt list per run (artifact `broad-coverage-ranking-<sha>`)
- `governance/coverage/criticality_manifest.json`: user-directed high-criticality files (B10 cluster)
- Targeted tests in `tests/coverage/` for phase 1, 4, 8, execution tracker, debug pipeline
- MIP B0 canonical coverage evidence with per-test contexts (`dynamic_context = test_function`)
- #1859 proposes a PR-level new-debt ratchet

MISSING

- No floor that rises: the gate is a fixed 75%, red on every PR
- No diff coverage on new or changed lines
- No wave plan with owners and dates
- 34 one-shot or legacy files sit in the denominator without a disposition
- Per-test contexts are recorded but not used to pick impacted tests (N05 pre-match)
- Coverage is not in the iteration census, so trends are invisible

COVERAGE · WAVE SEQUENCE (PROPOSED)

Six waves from 38% to 80%, each one raising the floor



Line coverage, simulated from today's per-file data: each wave raises its files to the stated level. The floor only moves up after exact-main CI confirms the new value (trigger T16). W1 changes the denominator, so it needs a written disposition per file.

COVERAGE · WAVE TARGETS

Which files each wave works on

WAVE	FILES	SIZE	NOTE
W1	abacus_v21_* (17), deploy_* and run_* (6), demo, ci_monitor_local, clone_based_validator, _LEGACY_phase0_setup and others	34 files · 5,017 statements	Each file gets keep+test, archive or omit, in writing. DOW_DEPLOYMENT_ORCHESTRATOR.py is a keep-and-test candidate
W2	phase9 · phase4 · phase1 · phase8 · phase7 · phase6 · phase2 · phase3 · phase5 · phase0_init	1,749 missing lines	Extend tests/coverage/ phase tests; they already exist for 1, 4, 8
W3	agents/self_improvement_agent · convergence/maturity_tracker · core/canonical_index · core/link_tracker · generators/master_reconciliation · integrations/git_manager …	53 modules · 4,382 missing	DMAIC_V3 root 1,842 · generators 646 · core 591 · convergence 504 · agents 385
W4	Partly covered DMAIC_V3 modules (debug_pipeline, execution_tracker …)	34 modules	Depth work: branches and error paths
W5	tools, scripts, models, src/dmaic, src/qplant_presentation_engine, renderers, integration	136 files	Mostly depth; renderers and integration have no tests yet

COVERAGE IN THE PROTOCOL

Where coverage sits in the lanes, nodes and triggers

AREA \ WAVE	W0	W1	W2	W3	W4	W5
DMAIC_V3	guard	guard	work	work	work	
repo root	guard	work				
tools	guard					work
scripts	guard					work
models	guard					work
src/*	guard					work
renderers, integration	guard					work

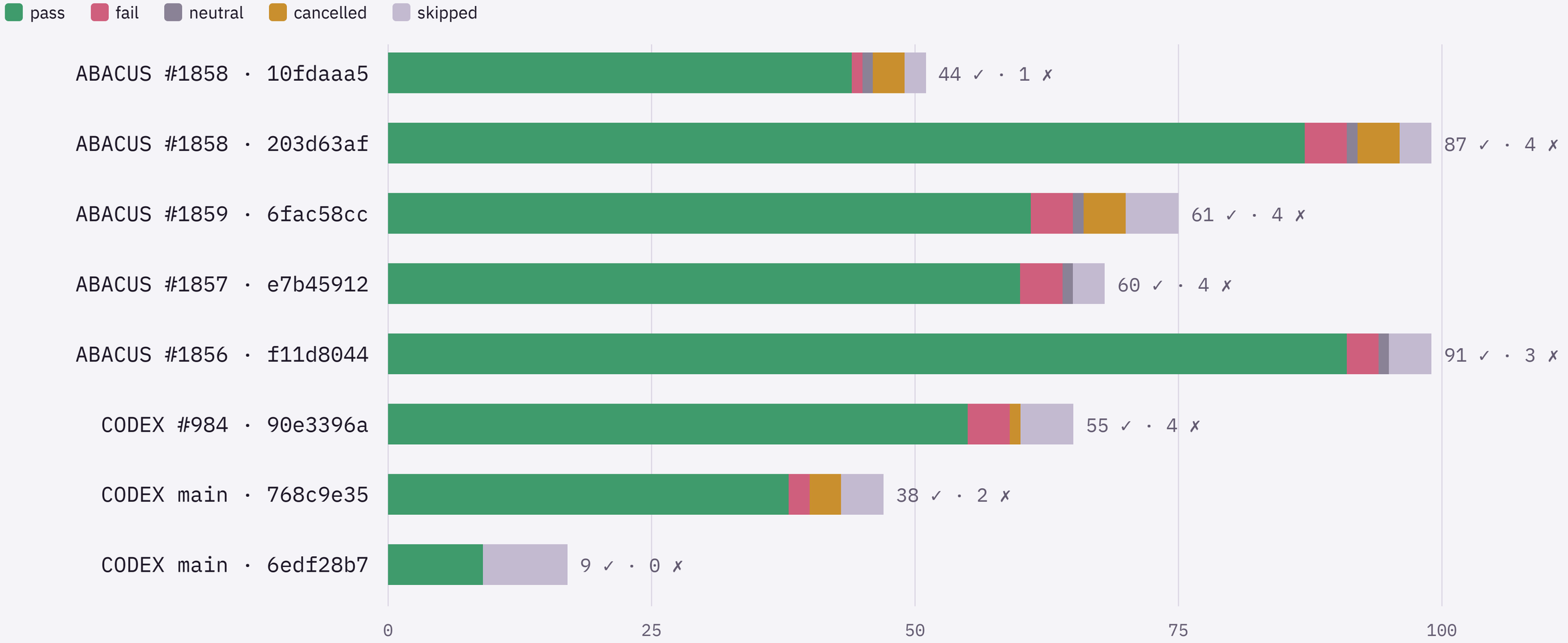
guard = ratchet and diff coverage protect the area; work = the wave adds tests or a disposition.

ATTACHMENT POINTS

LANE	New **Coverage** lane in tier Verify, owned by ABACUS `ci-cd-tests` and MIP B0
N04	Worker lane measures line + branch per test
N05	Pre-match uses per-test contexts to run only impacted tests
N06	Fixed-threshold red = inherited debt; diff coverage below 80% = source defect
N14	Exact-main CI records total, breadth, depth, per area
N12 · CENSUS	One coverage row per iteration; trend read back each pulse
T16 (NEW)	Main measured ≥ floor + 1 → raise the floor. Guard: exact-main CI green

CENSUS

Check outcomes per head, to scale



RUNTIME · VERSION CONTROL · DEBUG

What ran, where, on which commit

VERSION CONTROL	
ABACUS #1858	fix/gate-owner-attestation-fallback @ 203d63af
ABACUS MAIN	53688f26 · #1838 · 2026-10-06
CODEX MAIN	6edf28b7 · 2026-10-12 14:40 CEST
CODEX #984	merge 768c9e35 · 12 commits
RULESET	ABACUS 24494381 · active · 0 bypass

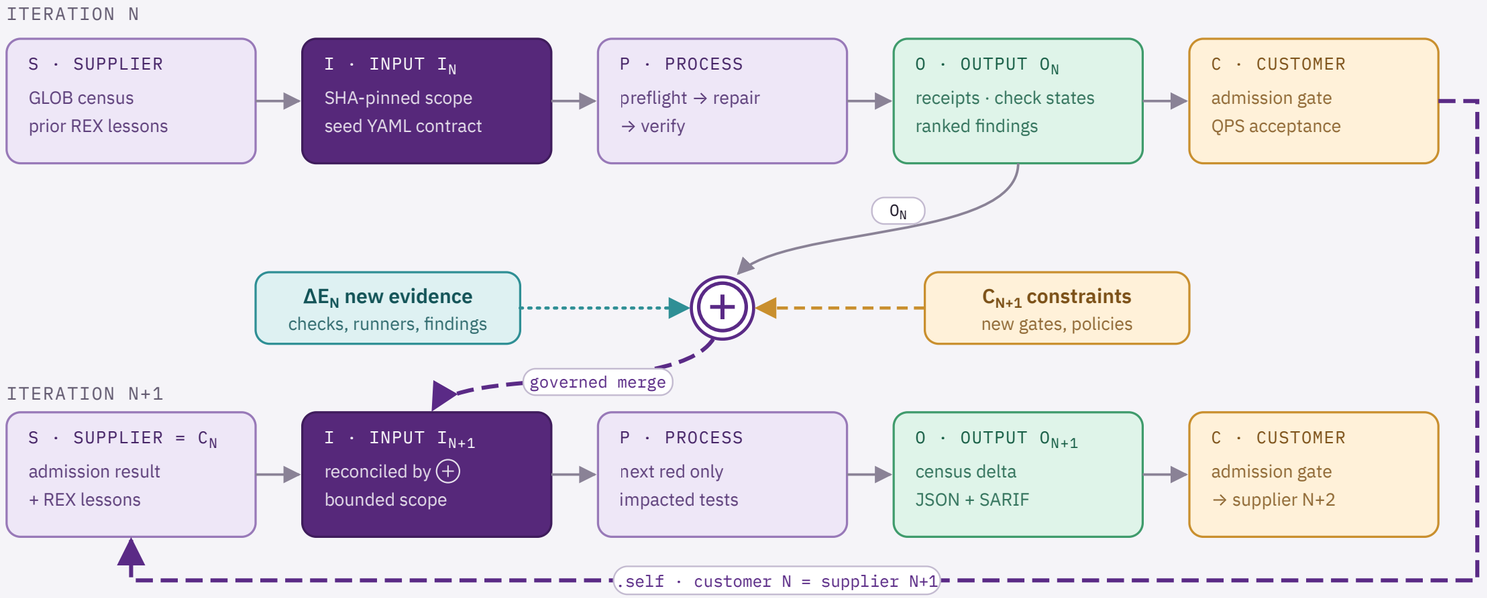
RUNTIME AND DEBUG	
PYTHON	3.10 · 3.11 · 3.12 (ABACUS CI matrix, all green on #1858)
RUNNERS	CPython (GitHub-hosted runners) · ubuntu-24.04
DEBUG LANES	CODEX main: Docker real runtime LLDB, live lldb-dap + MCP, live debugpy + js-debug — all green
LOCAL CHECK	Python 3.13: 61 governance tests pass; inventory check passes after regeneration
NOT YET	.self harness census, Docker smoke, live port probing

RECURSIVE SIPOC

The customer of N supplies N+1

$$I_{N+1} = O_N \oplus \Delta E_N \oplus C_{N+1}$$

O_N	last output: receipts, check states, ranked findings
ΔE_N	new evidence: completed checks, runner behaviour
C_{N+1}	new constraints: gates, policies, review rules
$\oplus$	governed reconciliation, never plain concatenation



LANES

Nine lanes, four tiers, existing owners

Code Purple

Discover	Verify	Analyse and remember	Authorise and deliver
Global lookup GLOB.yaml · mesh census CODEX	**Runtime and self-test** scripts/testpilot_preflight.py ABACUS	**Derived analysis** DOW ABACUS	**QPS domain** cryoplant-project sources cryoplant-project
Dock interface .mesh/MESH_DOCK.yaml mesh	**Lint and configuration** pre-commit · Ruff · YAML/JSON/TOML both	**Semantic and provenance** KEB · REX CODEX	**Delivery** Actions · PR Feedback Gate · Pages GitHub
	Security Bandit · Semgrep · SARIF both		

Runtime dock ladder: ABACUS devcontainer ports 8000 · 8080 · 5000 and the CODEX compose validator are **configured**; listening, published and probed are unproven.

BURNDOWN

What is left, who owns it, by when

#	ITEM	STATUS	OWNER	TARGET
1	**Land ABACUS #1858**	in progress	GBO + Claude	13 Oct
2	**Hardening #1878, #1879**	fixed in PR	verify after merge	post-merge
3	**CODEX post-merge reds**	classified	validate fixed · OSV inherited	done
4	**CODEX admission ruleset**	decision	GBO	14 Oct
5	**Classify recurring reds**	confirmed	4 inherited from main	done
6	**Runnable .self harness**	todo	Claude	16 Oct
7	**Scheduled readback pulse (T12)**	todo	Claude	16 Oct
8	**Docker smoke, port probing**	not built	Claude	23 Oct
9	**Cross-repo adoption (T15)**	gated on 1	—	after #1858

APPENDIX

Edge reference E01–E19

ID	FROM → TO	TYPE	MEANING
E01	N01→N02 Mission controller → Asymmetric allocation	Sequence	Every pulse starts at the controller.
E02	N02→N04 Asymmetric allocation → Parallel worker lanes	Sequence	Costly checks get dedicated lanes.
E03	N02→N03 Asymmetric allocation → Async I/O	Sequence	Cheap I/O runs without blocking the workers.
E04	N04→N05 Parallel worker lanes → Preflight + pre-match cache	Sequence	Per-lane JSON and SARIF results.
E05	N03→N05 Async I/O → Preflight + pre-match cache	Evidence	Prior receipts offered for exact-match reuse.
E06	N05→N06 Preflight + pre-match cache → First completed material red?	Sequence	Merged preflight result.
E07	N06→N07 First completed material red? → Isolated repair	Material red	A completed material red exists.
E08	N07→N08 Isolated repair → Impacted retest	Material red	Repair committed on an isolated branch.
E09	N08→N09 Impacted retest → Exact-head final admission	Sequence	Retest receipt goes to exact-head verification.
E10	N06→N09 First completed material red? → Exact-head final admission	Pass	No material red among completed checks.
E11	N09→N10 Exact-head final admission → All non-waivable gates green?	Sequence	Check and review states on the exact head.
E12	N10→N11 All non-waivable gates green? → HOLD	Hold	A non-waivable gate is red or pending.
E13	N10→N13 All non-waivable gates green? → Single-flight merge controller	Pass	Every non-waivable gate is green on the same SHA.
E14	N13→N14 Single-flight merge controller → Exact-main CI · metrics	Pass	One PR merged at a time.
E15	N14→N15 Exact-main CI · metrics → Output N · control	Pass	Main green; metrics and publication read back.
E16	N11→N12 HOLD → REX · KEB evidence	Evidence	Blocking reason recorded.
E17	N12→N03 REX · KEB evidence → Async I/O	Evidence	Lessons and receipts reused in the next pulse.
E18	N11→N01 HOLD → Mission controller	Hold	A material change triggers the next pulse.
E19	N15→N01 Output N · control → Mission controller	.self recursion	Validated output becomes input N+1 through ⊕.

SOURCES

Where the figures come from

- ABACUS #1858 · checks, gate output, review threads (GitHub API, 17:27)
- ABACUS #1878 and #1879 · follow-up hardening issues
- #1859, #1857, #1856 · check-run tallies
- CODEX #984 · merge record and check runs; CODEX main 768c9e35 and 6edf28b7
- ABACUS ruleset 24494381 (Main Hard Gate) · `.github/workflows/pr-feedback-gate.yml`, `ci/governance/pr_feedback_gate.py`
- Code Purple execution seed · `federation/mesh/CODE_PURPLE_EXECUTION_SEED_20261012.yaml`

LIFT · 2026-10-18 13:27 UTC READBACK

Not lifted: 2 of 5 criteria met

OPEN ABACUS PRS

20 → 13

baseline at declaration → now

MERGES SINCE #1858

97

gate green on 0; 97 with red checks

COVERAGE (LINE + BRANCH)

42.6%

recorded 15 Oct on main 803700a3; from 36.9%

RULESET 24494381

incomplete

bypass=6, rules=3; missing: required_status_checks rule absent

ID	LIFT CRITERION	MET	EVIDENCE
L1	**Unblocker PR merged**	✓	merged_at=2026-10-12T16:17:07Z
L2	**Gate restored (ruleset active, no bypass, expected rules present)**	✗	active, bypass=6, rules=3; missing: required_status_checks rule absent
L5	**Admission integrity (no merge since the unblocker without a green gate)**	✗	97 merged; gate green on 0; 97 merged with red checks
L3	**Merge flow restored (at least 5 merges to the default branch since the unblocker)**	✓	97 merges since 2026-10-12T16:17:07Z
L4	**Open PR backlog at most a quarter of the baseline**	✗	13 open (baseline 20)

The 13 Oct check passed L2 falsely: it read enforcement and bypass actors but not rule content. Fixed (EVR-ABACUS-0001). Follow-ups #1878 and #1879 verified on main. Red nightly on main: Code Quality Checks, Full Red Proof Summary, Security Scanning, Unit Tests (Python 3.10, macos-latest), Unit Tests (Python 3.10, ubuntu-latest), Unit Tests (Python 3.10, windows-latest), Unit Tests (Python 3.11, macos-latest), Unit Tests (Python 3.11, ubuntu-latest), Unit Tests (Python 3.11, windows-latest), Unit Tests (Python 3.12, macos-latest), Unit Tests (Python 3.12, ubuntu-latest), Unit Tests (Python 3.12, windows-latest).

AFTER-ACTION

What worked, what to change next time

WORKED

- **Readback before action.** It showed #1858 could not admit itself (the gate runs main's code) and that #984 had merged red.
- **Classify against main.** Four reds were red in main's nightly run, so accepting them under break-glass rested on evidence.
- **Smallest fix.** A 3-line regenerated doc cleared the only red the PR caused.
- **Owner / agent split.** Owner: bypass, merge, restore. Agent: threads, issues, fix, readback.
- **Measurable lift.** Criteria read from GitHub; one of them was too weak (below).

CHANGE

- **Break-glass deleted the rules instead of adding a bypass, and the lift check did not read rule content.** 61 PRs merged ungated. Use a bypass actor; L2 now compares rules; L5 audits every merge.
- **Snapshots went stale twice in an hour.** Run the readback every pulse and on a schedule.
- **A decision was written against an old head.** Threads were resolved as "residual risk" minutes after fixes were pushed. Re-read the head before writing any decision.
- **The seed repo has no gate.** CODEX #984 merged with 4 reds. Decide on a CODEX ruleset.
- **One reviewer was a single point of failure.** The owner-attested fallback is now on main; test it on the next outage.
- **Inherited reds stay red.** Coverage W0 floor and diff coverage due 16 Oct.

REUSABLE METHOD

Code Purple in eight steps

STEP	NAME	ENTRY	ACTION	WHO
0	**Declare**	Merge flow stalled, the same checks red across PRs, or a required reviewer unavailable	Write incident.yaml: repos, gate, tracked PRs, owner, lift criteria	Owner
1	**Read back**	incident.yaml exists	Run readback.py; no action before the first census	Agent
2	**Classify every red**	Census on exact heads	Inherited (also red on main nightly) · source defect · review capacity · runner or infra	Agent
3	**Find the unblocker**	Classes known	Smallest change that lets the gate admit again; prove it cannot admit itself	Agent + owner
4	**Repair source defects only**	Unblocker chosen	Fix on the unblocker branch, impacted retest, exact-head readback	Agent
5	**Admit or break glass**	Only inherited or capacity reds left	Owner writes residual risk, adds a one-time bypass actor (never deletes rules), merges, removes the bypass, re-reads rule content	Owner
6	**Lift check**	Unblocker merged	readback.py evaluates L-criteria; census row saved	Agent
7	**Close out**	Lifted	Verify follow-ups on main and close; REX lesson; inherited reds become dated burn-down items	Agent + owner

Rule that never bends: the owner alone breaks glass, once, in writing, and restores the gate immediately. Nothing that changes a gate approves itself.

ADAPTING IT

One file to adapt; the same steps elsewhere

INCIDENT.YAML KNOBS

`repos[].ruleset_id, gate_check`	Which gate is blocked and which single check it requires
`repos[].nightly_workflow`	The scheduled run on the default branch that defines the inherited set
`tracked_prs[].role`	unblocker · debt-ratchet · queued · seed: decides lift and report order
`lift_criteria[]`	pr_merged · ruleset_clean · merges_since · open_prs_at_most, with thresholds
`follow_up_issues[].check`	What to verify on main before closing each follow-up
`baseline`	Numbers at declaration, for the before/after census

CONCEPT	CODE (THIS CASE)	ENGINEERING DOCUMENTS	DATA PIPELINE
Gate	Required CI check	Approval signature on a controlled document	Validation step before a dataset is published
Inherited red	Red on main nightly	Finding already open in the register	Check already failing on the last good release
Unblocker	PR that restores admission	Revision that restores reviewability	Fix that lets validation run again
Break-glass	One-time ruleset bypass	Documented concession, owner-signed	Manual release with sign-off
Lift	Merges flow, backlog down	Reviews close, queue down	Releases resume

Only the code column has a working readback today; the other two are mappings for reuse.

TOOLKIT

What runs the method

FILE	DOES
kit/incident.yaml	The instance: repos, gate, tracked PRs, owner, baseline, lift criteria. The only file to edit.
kit/readback.py	Reads GitHub through `gh`; classifies reds against the main nightly run; checks lift criteria; writes `census.json` and `after_action.md`.
index.html	The interactive page: pulse graph, lifecycle triggers, census, admission board.
build_deck.js · merge_deck.py	This 16:9 deck and its bookmarks, from the same model.
code-purple.md	Markdown parent with the same anchors as the PDF links.

Run: `python kit/readback.py kit/incident.yaml` → `kit/out/census.json`, `kit/out/after_action.md`. Lifted when every L-criterion is met.